kilometers. Kares maneuvers the starship into that orbit, and your crew begin their telescopic, photographic observations.

Over the next six days the two holes gradually move closer to each other and speed up their orbital motion. One day before coalescence, the distance between them has shrunk from 30,000 to 18,000 kilometers and their orbital period has decreased from 13 to 6.3 seconds. One hour before coalescence they are 8400 kilometers apart and their orbital period is 1.9 seconds. One minute before coalescence: separation 3000 kilometers, period 0.41 second. Ten seconds before coalescence: separation 1900 kilometers, period 0.21 second.

Then, in the last ten seconds, you and your starship begin to shake, gently at first, then more and more violently. It is as though a gigantic pair of hands had grabbed your head and feet and were alternately compressing and stretching you harder and harder, faster and faster. And then, more suddenly than it started, the shaking stops. All is quiet.

"What was that?" you murmur to DAWN, your voice trembling.

"Tikhii, tikhii," she replies soothingly. "That was the undulating tidal force of gravitational waves from the holes' coalescence. You are accustomed to gravitational waves so weak that only very delicate instruments can detect their tidal force. However, here, close to the coalescing holes, they were enormously strong—strong enough that, had we parked our starship in an orbit 30 times smaller, it would have been torn apart by the waves. But now we are safe. The coalescence is complete and the waves are gone; they are on their way out into the Universe, carrying to distant astronomers a symphonic description of the coalescence."[36]

Training one of your crew's telescopes on the source of gravity below, you see that DAWN is right, the coalescence is complete. Where before there were two holes there now is just one, and it is spinning rapidly, as you see from the swirl of infalling atoms. This hole will make an ideal power generator for your crew and thousands of generations of their descendants.

By measuring the starship's orbit, Kares deduces that the hole weighs 45 solar masses. Since the parent holes totaled 48 solar masses, 3 solar masses must have been converted into pure energy and carried off by the gravitational waves. No wonder the waves shook you so hard!

As you turn your telescopes toward the hole, a small object unexpectedly hurtles past your starship, splaying brilliant sparks profusely in all directions, and then explodes, blasting a gaping hole in your ship's